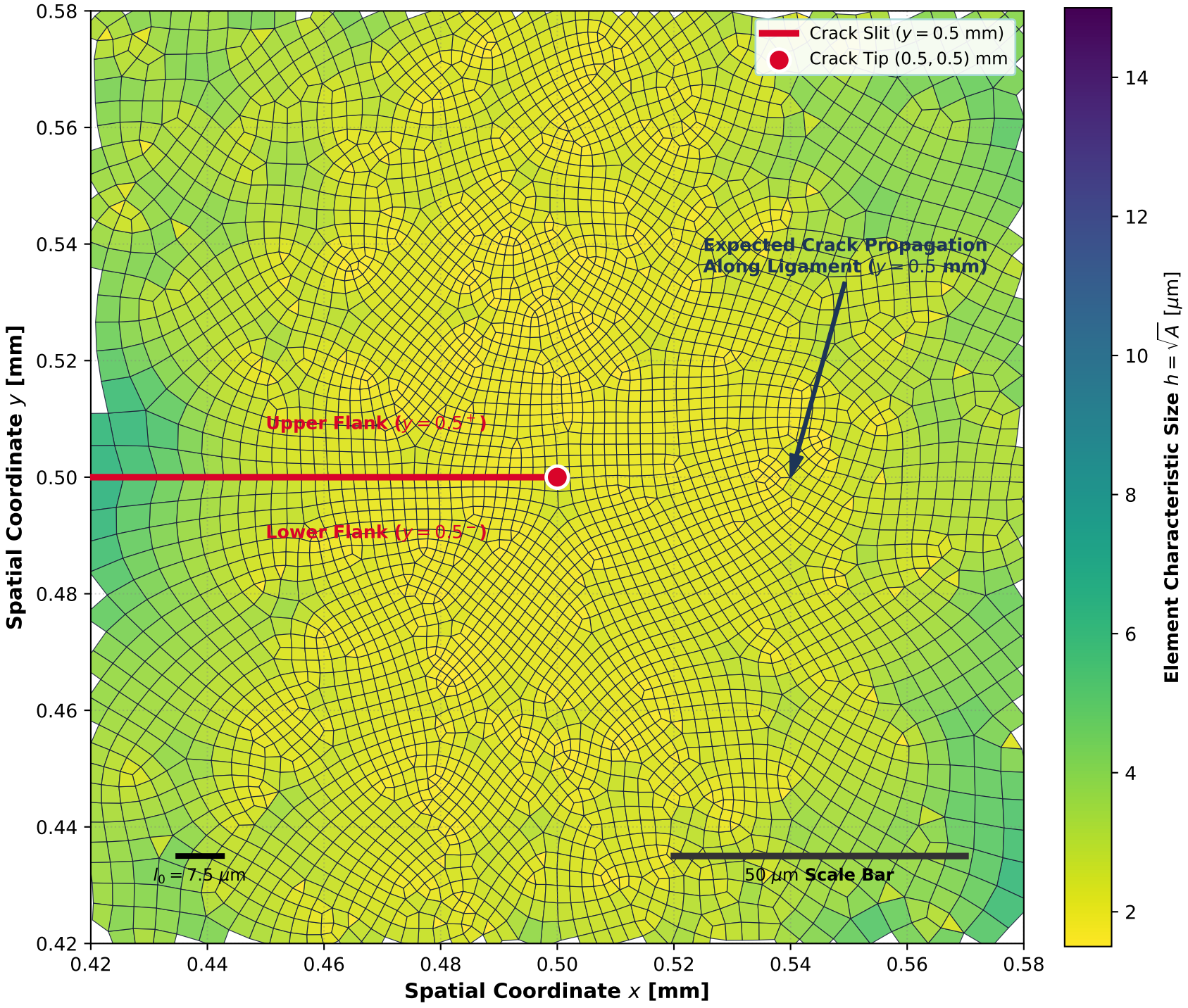

Crack-Tip Refinement Corridor & Element Sizing Resolution
(Tip Resolution $h_{tip} \approx 1.91\ \mu m$, Phase-Field Regularization $l_0 = 7.5\ \mu m$)
Crack Slit ($y = 0.5$ mm)
Crack Tip (0.5, 0.5) mm
Expected Crack Propagation
Along Ligament ($y = 0.5$ mm)
Upper Flank ($y = 0.5^+$)
Lower Flank ($y = 0.5^-$)
$l_0 = 7.5\ \mu m$
50 μm Scale Bar
Spatial Coordinate y [mm]
Spatial Coordinate x [mm]
Element Characteristic Size $h = \sqrt{A}$ [μm]
0.42
0.44
0.46
0.48
0.50
0.52
0.54
0.56
0.58
2
4
6
8
10
12
14